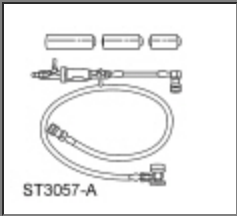


Refrigerant Oil Adding

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

	R-134a Loop/Add On Injector Kit-Set 219-703710 or equivalent
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Material

Item	Specification
Motorcraft® PAG Refrigerant Compressor Oil YN-12-D	WSH-M1C231-B

Refrigerant Oil Adding

NOTICE: During normal A/C operation, oil is circulated through the system with the refrigerant, and a small amount is retained in each component. If certain components of the system are removed, some of the refrigerant oil will go with the component. To maintain the original total oil charge, it is necessary to compensate for the oil lost by adding oil to the system with the new part.

1. Refer to the chart below for refrigerant oil adding amounts when new components are installed.

Component	PAG (polyalkylene glycol) Refrigerant Compressor Oil (YN-12-D) Oil Amount (fl oz (ml))	Method of Adding
After installing the A/C Compressor.	Service compressor contains full system charge of oil. Oil matching required plus amount collected during recovery.	Add directly to A/C compressor before installation
After installing the receiver/drier.	2 fl oz (60 ml) plus the amount collected during refrigerant recovery.	Inject to low-side service port during system charging.
After installing the receiver/drier element.	1 fl oz (30 ml) plus the amount collected during refrigerant recovery.	Inject to low-side service port during system charging.
After installing the condenser core.	1.5 fl oz (45 ml) plus the amount collected during refrigerant recovery.	Add directly to inlet port or inject to low-side service port during system charging.
After installing the condenser core, and the receiver/drier element	2 fl oz (60 ml) plus the amount collected during refrigerant recovery.	Add directly to inlet port or inject to low-side service port during system charging.
After installing the evaporator core.	1.5 fl oz (45 ml) plus the amount collected during refrigerant recovery.	Add directly to inlet port or inject to low-side service port during system charging.

Component	PAG (polyalkylene glycol) Refrigerant Compressor Oil (YN-12-D) Oil Amount (fl oz (ml))	Method of Adding
After installing/reassembling the thermostatic expansion valve.	Amount collected during refrigerant recovery.	Inject to low-side service port during system charging.
After installing a refrigerant Hose/Line.	2 fl oz (60 ml) plus the amount collected during refrigerant recovery ^a	Inject to low-side service port during system charging.
After an O-ring Leak Repair.	1 fl oz (30 ml) plus the amount collected during refrigerant recovery ^b	Inject to low-side service port during system charging.
After a service port leak repair.	Amount collected during refrigerant recovery.	Inject to low-side service port during system charging.

^aIf an excessive amount of PAG (polyalkylene glycol) Refrigerant Compressor Oil YN-12-D is lost due to a hose rupture/separation or other damage, the total system PAG (polyalkylene glycol) Refrigerant Compressor Oil YN-12-D capacity must be added. Also, it is necessary to replace the receiver/drier.

^bThe amount specified may be used for one or multiple O-ring leak repairs. Do not multiply the PAG (polyalkylene glycol) Refrigerant Compressor Oil YN-12-D amount by the number of O-ring leaks being repaired.

Compressor Type(s)

NOTICE: An A/C refrigerant analyzer must be used before the recovery of any vehicle's A/C refrigerant. Failure to do so puts the shop's bulk refrigerant at risk of contamination. If the vehicle's A/C refrigerant is contaminated, refer the customer to the service facility that carried out the last A/C service. If the customer wishes to pay the additional cost, use the A/C recovery equipment that is designated for recovering contaminated A/C refrigerant. Dispose of all contaminated A/C refrigerant as hazardous waste. For all equipment, follow the manufacturer's instructions.

NOTICE: Motorcraft® PAG Refrigerant Compressor Oil YN-12-D only must be used as a refrigerant system lubricant. Addition of any oil other than Motorcraft® PAG Refrigerant Compressor Oil YN-12-D to the refrigerant system will damage the A/C compressor and contaminate the refrigerant system.

NOTICE: During normal A/C operation, oil is circulated through the system with the refrigerant, and a small amount is retained in each component. If certain components of the system are removed, some of the refrigerant oil will go with the component. To maintain the original total oil charge, it is necessary to compensate for the oil lost by adding oil to the system with the new part.

NOTICE: Oil balance must be performed when replacing a compressor where system flush will not be performed.

NOTE: *The purpose of oil balance is to ensure that the amount of oil in the system is not increased or decreased as a result of the replacement component(s).*

NOTE: *New compressors come fully charged with Motorcraft® PAG Refrigerant Compressor Oil YN-12-D. The old compressor is only partially charged because some of the oil is elsewhere in the A/C system.*

1. Verify location of the drain plug by compressor type. Once compressor type has been identified, procedure to appropriate compressor type below for compressor oil matching steps.

2. **NOTICE:** Make sure the equipment is clean and free of foreign material.

NOTE: *During normal A/C operation, oil is circulated through the system with the refrigerant, and a small amount is retained in each component. If certain components of the system are removed, some of the PAG (polyalkylene glycol) oil remains with the component. To maintain the original total oil charge, it is necessary to compensate for the oil lost by adding oil to the system with the new part.*

Fill the system with the calculated quantity of new refrigerant oil in accordance with the A/C service unit manufacturer instructions.

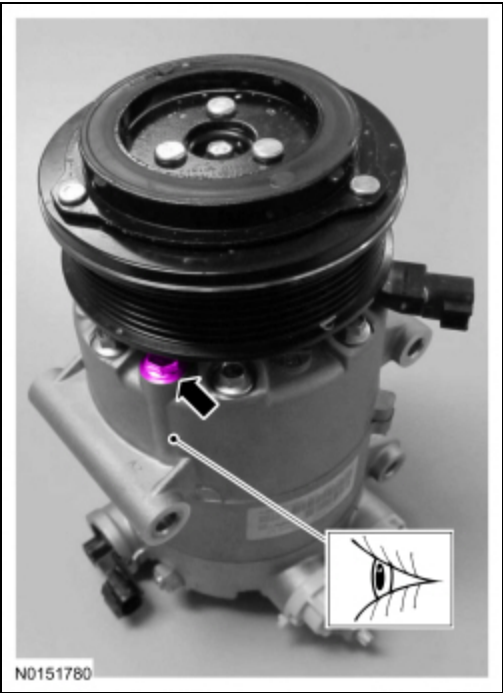
Type 1: (IVDC) Internal Variable Displacement Air Conditioning (A/C) Compressor

1.



Type 2: (EVDC) External Variable Displacement Air Conditioning (A/C) Compressor

1.



Type 3: (EVDC) External Variable Displacement Air Conditioning (A/C) Compressor

1.



Type 4: Fixed Air Conditioning (A/C) Compressor

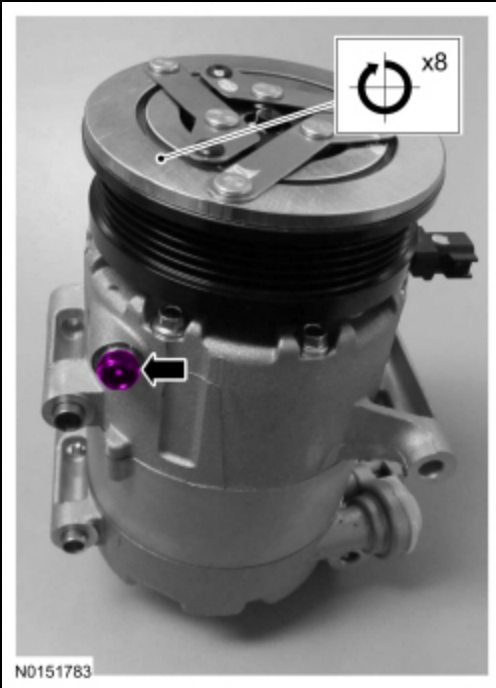
1.



Type 1

NOTICE: Make sure the old A/C compressor shaft is rotating and not just the A/C pulley.

1. Remove the oil drain bolt and rotate the old A/C compressor clutch 6 to 8 revolutions while collecting oil in a clean measuring device.

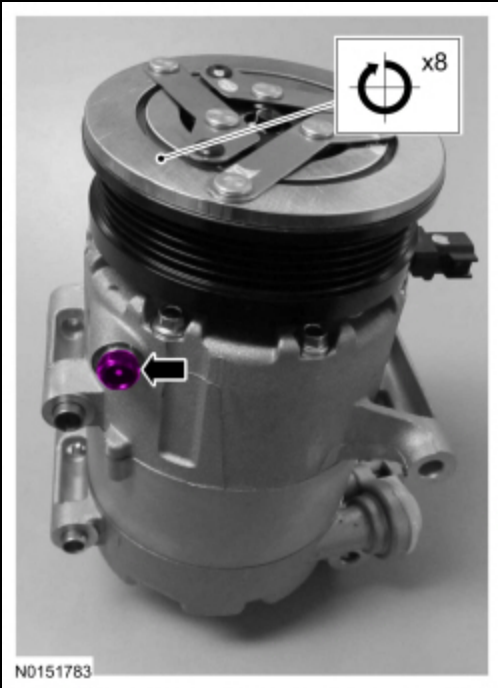


2. Install the oil drain bolt in the old A/C compressor.
 - To install, tighten to 20 Nm (177 lb-in).



3. Remove the suction and discharge port seal cap from the new A/C compressor.
4. **NOTICE: Make sure the new A/C compressor shaft is rotating and not just the A/C pulley.**

Remove the drain bolt and rotate the new A/C compressor clutch 6 to 8 revolutions while collecting oil in a clean measuring device.



5. Measure the amount of oil drained for the old A/C compressor and add the same amount of new oil to the new A/C compressor.
6. Install the oil drain bolt in the new A/C compressor.
 - To install, tighten to 20 Nm (177 lb-in).



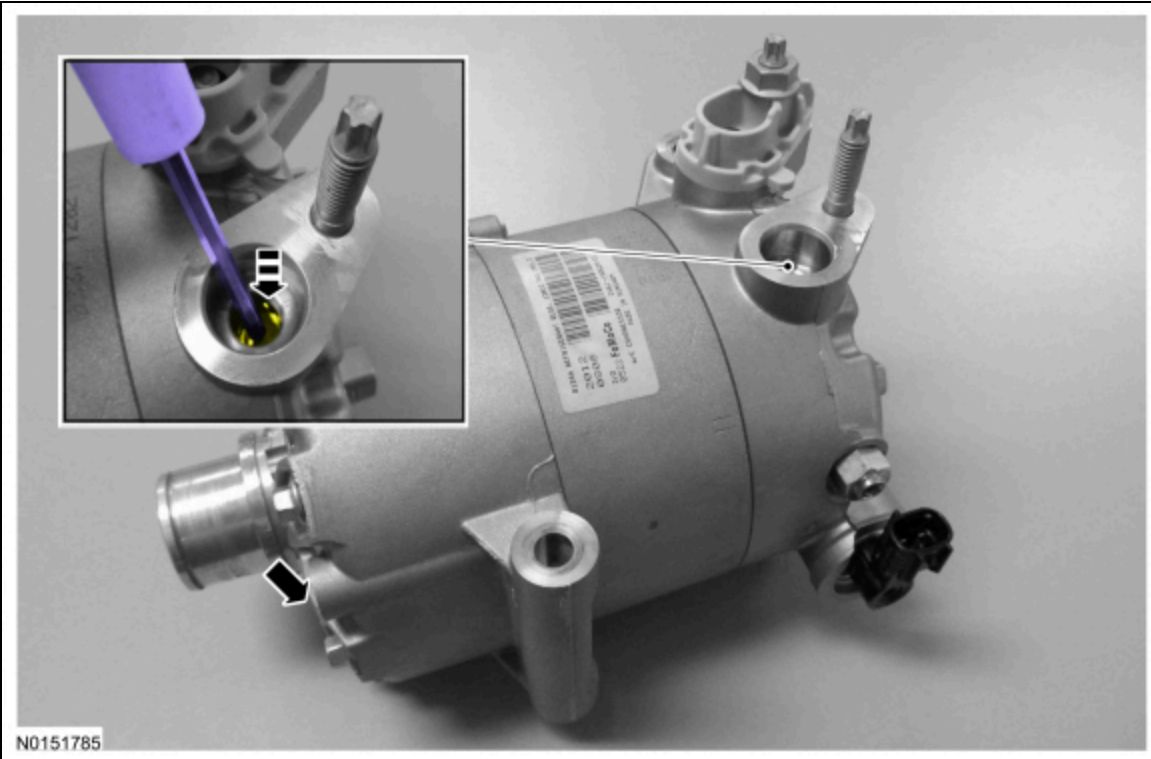
7. Install the suction and discharge port seal cap from the new A/C compressor.

Type 2

1. Remove the A/C Clutch and A/C Clutch Field Coil from the old A/C compressor. For additional Information, refer to Section 412-01.
2. Remove the oil drain bolt from the old A/C compressor.



3. Depress the suction damper to allow proper ventilation while draining oil from the old A/C compressor.



4. Drain as much oil from the old A/C compressor.

5. Install the oil drain bolt in the old A/C compressor.

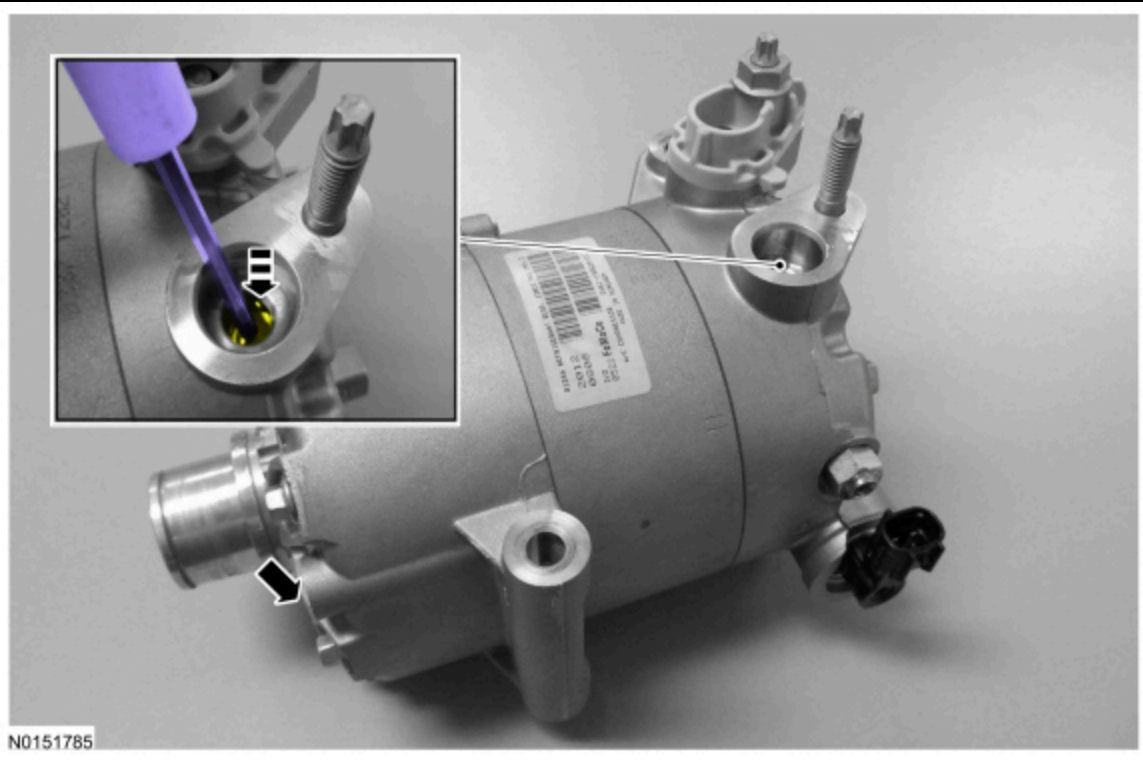
- To install, tighten to 9 Nm (80 lb-in).



6. Remove the A/C Clutch and A/C Clutch Field Coil from the new A/C compressor. For additional information, refer to Section 412-01.
7. Remove the oil drain bolt from the new A/C compressor.



8. Remove the suction port seal cap.
9. Depress the suction damper to allow proper ventilation while draining oil from the old A/C compressor.



10. Drain as much oil from the new A/C compressor.

11. Measure the amount of oil drained for the old A/C compressor and add the same amount of new oil to the new A/C compressor.

12. Install the oil drain bolt in the new A/C compressor.

- To install, tighten to 9 Nm (80 lb-in).



13. Install the suction port seal cap.

14. Install the A/C Clutch and A/C Clutch Field Coil on the new A/C compressor. For additional information, refer to Section 412-01.

Type 3

NOTICE: Make sure the old A/C compressor shaft is rotating and not just the A/C pulley.

1. Remove the oil drain bolt and rotate the old A/C compressor clutch 6 to 8 revolutions while collecting oil in a clean measuring device.



2. Install the oil drain bolt in the old A/C compressor.
 - To install, tighten to 30 Nm (22 lb-ft).



3. Remove the suction and discharge port seal cap from the new A/C compressor.
4. **NOTICE: Make sure the new A/C compressor shaft is rotating and not just the A/C pulley.**

Remove the drain bolt and rotate the new A/C compressor clutch 6 to 8 revolutions while collecting oil in a clean measuring device.



5. Install the oil drain bolt in the old A/C compressor.
 - To install, tighten to 30 Nm (22 lb-ft).



6. Remove the suction and discharge port seal cap from the new A/C compressor

7. **NOTICE: Make sure the new A/C compressor shaft is rotating and not just the A/C pulley.**

Remove the drain bolt and rotate the new A/C compressor clutch 6 to 8 revolutions while collecting oil in a clean measuring device



8. Measure the amount of oil drained for the old A/C compressor and add the same amount of new oil to the new A/C compressor.
9. Install the oil drain bolt in the new A/C compressor.
 - To install, tighten to 30 Nm (22 lb-ft).



10. Install the suction and discharge port seal cap on the new A/C compressor.

Type 4

NOTICE: Make sure the old A/C compressor shaft is rotating and not just the A/C pulley.

1. Rotate the old A/C compressor clutch 6 to 8 revolutions while collecting oil in a clean measuring device from the A/C compressor discharge port.



2. Remove the suction and discharge port seal cap from the new A/C compressor.

3. **NOTICE: Make sure the new A/C compressor shaft is rotating and not just the A/C pulley.**

Rotate the new A/C compressor clutch 6 to 8 revolutions while collecting oil in a clean measuring device from the A/C compressor discharge port.



4. Measure the amount of oil drained for the old A/C compressor and add the same amount of new oil to the new A/C compressor.

5. Install the suction and discharge port seal cap on the new A/C compressor.

Oil Injection Using a Dye/Lubricant Injector

NOTE: Fluorescent refrigerant system dye is added to the refrigerant system at the factory to assist in refrigerant system leak diagnosis using a Rotunda-approved UV black light. It is not necessary to add additional dye to the refrigerant system before diagnosing leaks, even if a significant amount of refrigerant has been removed from the system. Replacement suction accumulators and receiver/driers are shipped with a fluorescent dye wafer included in the desiccant bag which dissolves after approximately 30 minutes of continuous A/C operation. It is not necessary to add dye after flushing the refrigerant

system because a new suction accumulator or receiver/drier is installed as part of the flushing procedure. Replace hose/accumulator, receiver/drier or receiver/drier element.

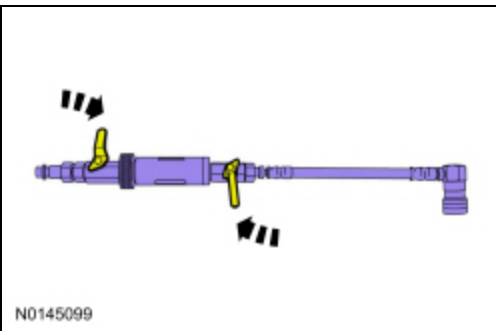
NOTE: The dye/lubricant injector is included as part of the 219-00069 R-134a Loop/Add On Injector Kit-Set.

NOTE: Before using the R-134a Loop/Add On Injector Kit-Set for the first time, refer to the equipment manufacturer's instructions on evacuation of non-condensable gases from the hoses.

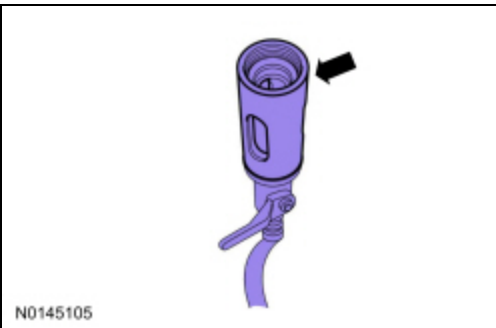
NOTE: If fluorescent leak detection dye is also to be added during A/C charging, the dye may be added to the dye/lubricant injector along with the refrigerant oil.

NOTE: Refrigerant system pressure should be between 413-551 kPa (60-80 psi) at 24° C (75° F) with the engine off and cool.

1. Evacuate the refrigerant system. For additional information, refer to Air Conditioning (A/C) System Recovery, Evacuation and Charging in this section.
2. Assemble the dye/lubricant injector using the correct adapters to match the amount of refrigerant compressor oil to be injected.
3. Verify that all the valves on the dye/lubricant injector are closed.



4. Fill the dye/lubricant injector with the correct amount of clean, new PAG (polyalkylene glycol) oil.



5. Install the dye/lubricant injector between the low-side service gauge port valve and the refrigerant service station or manifold gauge set.
6. **NOTE:** Make sure all tools and hoses are clear of the engine cooling fan and drive belt before starting the engine. Failure to keep tools and hoses clear from the engine cooling fan and drive belt results in damage to the tools and/or vehicle.

Open all valves and charge the refrigerant system. For additional information, refer to Air Conditioning (A/C) System Recovery, Evacuation and Charging in this section.